

Michael Baylard

Senior AI Engineer — Inference Platforms · Vector Search · RAG Infrastructure

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SUMMARY

Senior AI Engineer with 4 years building **production inference and retrieval infrastructure** at Fortune 100 scale. Deployed real-time classification on **A10 GPU clusters** processing **38M+ images** (F1 0.44 → 0.92) with model compression for constrained deployment. Architected a **multi-provider LLM serving layer** with multi-tier failover routing, per-API rate limiting (450–950 RPM), and ThreadPoolExecutor parallelism (5–50 workers per stage) across 8 API providers. Built **production RAG infrastructure on PostgreSQL/pgvector** with 35+ table multi-tenant schema, agency-scoped RBAC, and streaming inference via Vercel AI SDK v5. Comfortable owning systems from the **inference layer to schema to deploy** — schema governance across 76+ table schemas in 5 databases at John Deere.

TECHNICAL SKILLS

- Inference & Serving Infrastructure:** — GPU inference (A10 / g5 clusters), model serving, model compression (SwinTransformer → MobileNetV2), multi-tier failover routing (MiniMax → OpenRouter → Groq), per-API rate limiting, ThreadPoolExecutor parallelism (5–50 workers per stage), streaming inference (Vercel AI SDK v5), crash-safe incremental JSONL processing, thread-safe append-under-lock
- Vector Search & RAG:** — PostgreSQL/pgvector (production), embedding-based retrieval, DINOv2 embedding-based dataset expansion, multi-tenant RAG with agency-scoped RBAC, English-to-SQL co-agent with parallel tool-calling, structured-JSON extraction with multi-tier parse fallback, schema-driven prompts (15+ field business schema)
- LLM Orchestration:** — 6-provider routing (Claude Opus 4.7, MiniMax M2.7 Highspeed, Gemini 3.1 Pro, Groq Compound + Llama 3.3 70B, Exa, Tavily), Gemini native JSON mode, output guardrails, model evaluation, prompt engineering, 6 MCP servers integrated
- Production ML / Computer Vision:** — PyTorch, TensorFlow/Keras, scikit-learn, PyCaret, OpenCV, MobileNet, EfficientNet, SwinTransformer, DINOv2, FiftyOne (Voxel51), MLflow, embedding-based dataset curation
- Data & Platform:** — Delta Lake, Unity Catalog, schema governance, ETL architecture, Docker, CI/CD (GitHub Actions), AWS (Databricks), Azure, Vercel, FastAPI, Next.js API routes, query optimization
- Databases:** — PostgreSQL / Neon (pgvector, RLS, triggers, JSONB), Databricks (Delta Lake, Unity Catalog), schema design, migrations, partition strategies
- Languages:** — Python, TypeScript / JavaScript, SQL, Bash

PROFESSIONAL EXPERIENCE

- John Deere — AI Engineer (FurrowVision program)** Chicago, IL
2023 – Present | Fortune 100 · Precision Agriculture / Industrial AI
- Production GPU inference at scale**
- Deployed real-time classification pipeline on **A10 GPU clusters** processing **38M+ sensor images** — F1 improved from **0.44 → 0.92** through 8 months of iterative dataset curation, model compression (SwinTransformer → MobileNetV2), and DINOv2 embedding-based expansion
 - Curated 15,000-image training corpus from 38M+ raw images using **embedding-distance retrieval** — engineered self-serve workflow enabling 4,000+ images added post-handoff with zero data-science involvement
 - Validated pipeline across geographies — integrated Brazil sensor data with zero re-engineering
- Schema governance & platform infrastructure**
- Documented **76+ table schemas across 5 databases** (Unity Catalog generators, partition strategies, data lineage) — identified a 1B-row unpartitioned table and flagged 130M- and 42M-row tables for remediation; output consumed by all downstream teams
 - Defined **17 system metrics** with Systems Engineering, now standard across downstream analytics teams
 - Drove **3–5× labeling-throughput improvement** by leading SuperAnnotate → FiftyOne migration; delivered zero-follow-up platform handoff backed by 13 documentation files
- Forecasting & business impact**
- Built used-equipment forecast models with **company-best MAPE across 7 product lines** (PyCaret, Auto ARIMA) — analytics work influenced **\$82M+ in revenue across FY23–24**

2025 – Present | Productized AI service · Competitive Intelligence

- Built **multi-provider LLM serving layer** routing **8 API providers** with cost-optimized model selection: **MiniMax M2.7 Highspeed** (ICP scoring, quality review), **Gemini 3.1 Pro** (research synthesis, native JSON mode), **Claude Opus 4.7** (outreach generation), **Groq Compound + Llama 3.3 70B** (dossier research), **Tavily / Exa** (live-web retrieval), **Apollo** (org/people extraction), **OpenRouter** (failover routing)
- Engineered **multi-tier failover** (MiniMax → OpenRouter → Groq), **per-API rate limiting** (450–950 RPM), **ThreadPoolExecutor parallelism** (5–50 workers per stage), **crash-safe incremental JSONL** processing with thread-safe append-under-lock
- Implemented **structured JSON extraction** with multi-tier parse fallback (direct → code-block extraction → trailing-comma repair → regex object extraction) — schema-driven prompts (15+ field business schema, 10+ field contact schema), Gemini native JSON mode for guaranteed structured output
- Built **incremental result caching** + lead deduplication (domain normalization + email dedup) — JSONL-based scored-record dedup skips already-processed records, **reducing API cost 60–80%** on repeat client pipelines
- Orchestrated **16-stage pipeline** (Score → Research → Generate → Deliver) across 14+ Python scripts and 3 parallel execution tracks via config-driven multi-client architecture (context.json + YAML)
- Production scale: **scored 17,700+ companies, enriched 6,300+ contacts, 78% verified email rate, full pipeline in <3 hours** (was weeks of manual work)

Speaksense — Founding Engineer, RAG Platform

Chicago, IL

2024 – Present | Multi-tenant AI SaaS · YouTube Analytics

- Architected production **RAG system on PostgreSQL/pgvector** — English-to-SQL co-agent with parallel tool-calling, streaming inference via Vercel AI SDK v5, persistent chat memory
- Built **multi-tenant retrieval infrastructure: 35+ PostgreSQL tables**, agency-scoped RBAC (JSONB permissions), pgvector embeddings for semantic search, Stripe webhook-driven plan management
- Registered **8 agentic tools** (channel analysis, English-to-SQL, thumbnail evaluation, competitor intelligence, image generation, community trends)
- Integrated **6 MCP servers** (DB, docs, git, web search) — **~10× development velocity** across planning, implementation, deploy cycles
- **Stack:** Next.js 15, React 19, TypeScript, Neon PostgreSQL (pgvector), OpenAI GPT-4, Vercel AI SDK v5, Stripe, Docker

SELECTED PROJECTS

Multi-Provider Inference Platform — Klarix

Python, MiniMax M2.7 Highspeed, Gemini 3.1 Pro, Claude Opus 4.7, Groq Compound + Llama 3.3 70B, Exa, Tavily, Apollo API, OpenRouter, Playwright, PyMuPDF
8-provider routing with multi-tier failover; per-API rate limiting (450–950 RPM); ThreadPoolExecutor parallelism (5–50 workers per stage); crash-safe JSONL with thread-safe append-under-lock; 17,700+ companies scored at <3-hour wall-clock.

Production RAG on pgvector — Speaksense

Next.js 15, TypeScript, Vercel AI SDK v5, PostgreSQL/pgvector, OpenAI
35+ table multi-tenant schema with agency-scoped RBAC; English-to-SQL with parallel tool-calling; streaming chat UX with persistent memory; pgvector embeddings for semantic search.

Production GPU Inference Pipeline — John Deere

Python, PyTorch, TensorFlow/Keras, MobileNet, DINOv2, FiftyOne, Databricks (AWS)
38M+ images classified on A10 GPU clusters; F1 = 0.92 in production; model compression for constrained deployment; self-serve ML platform with embedding-distance retrieval for engineering teams.

Financial Forecasting System — John Deere

Python, PyCaret, Auto ARIMA, scikit-learn
Company-best MAPE across 7 product lines; engineered lag features and external economic signal correlation.

EDUCATION

B.S. Computer Science — Bradley University, Peoria IL · 2022 · Magna Cum Laude

Concentrations: Data Science · Cybersecurity

- **Capstone (Data Mining):** Built and deployed an ML system predicting student grade outcomes from prior coursework — authored the production prediction layer over a trained model

References, code samples, and additional project details available on request.